



Fig. 3: Soil moisture sensor (contact type)



Fig. 4: Motor pump

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

and the threshold value will be shown on the serial monitor in Arduino. Serial debugging is available in this program. Comment out if you do not wish to use the serial monitor.

Construction and testing

An actual-size, single-side PCB layout of the automated plant watering system is shown in Fig. 5 and its component layout in Fig. 6.

Assemble the components on the PCB to minimise errors. Alternatively, you can assemble them on a breadboard or Arduino prototyping shield or a general-purpose PCB. Upload the code to Arduino UNO board and

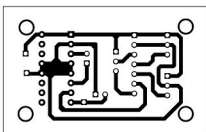


Fig. 5: Actual-size PCB layout of the circuit

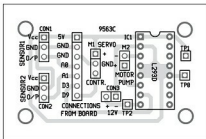


Fig. 6: Component layout of the PCB



Fig. 7: Installing water pump in the container



Fig. 8: Attaching the pipe on the servo horn

install the sensors in the soil of the potted plants. Do not immerse the sensors fully inside the soil.

Install the pump in a water container (refer Fig. 7) that can hold a few litres of water. Attach the water pipe on the servo motor horn as shown in Fig. 8.

Before powering the circuit on, you need to keep in mind the following macro definitions in the code:

1. Changing the angle of rotation of the servo horn toward the first pot and second pot. The default values are 70 degrees and 145 degrees.

2. Changing the watering time according to the size of the pot. The default values are five seconds and eight seconds.

3. Changing the threshold value according to your need. The default value is 600.

Place the flower pots where the pipe from the servo motor horn can easily reach them. When the moisture level dips below 600, servo horn rotates at an angle of 70 degrees. That is, after servo motor horn moves 70 degrees toward the first pot, the motor pump will be on for five seconds and then stop automatically. Then, the servo returns to its original position. Similarly, if you are using a second sensor, the servo motor horn will move to 145 degrees to the second biggest pot, motor pump will be on for eight seconds and then stop automatically. The servo returns to its original position.

Further application

Using the Arduino UNO board, you can water six different potted plants. By adding a few more lines in the code, you can water even more plants—by using the Arduino Mega 2560 board which has more analogue input pins.

You can also add an Ethernet or Wi-Fi shield and use the Twitter library, which will tweet from your plants side to send messages like: I need water, tank is empty, refill the tank, thanks for the water, and so on.

You can also add a 16×2 LCD to indicate moisture levels.

You can also enable the circuit to refill the tank after a few days, depending on the volume of the tank. ●

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